

SeungJun Tak

M.S.–Ph.D. Student · Intelligent Robotics · Sungkyunkwan University
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Education

- 2023.09 – Present

M.S.–Ph.D. Integrated Program

Sungkyunkwan University

Intelligent Robotics · Advisor: Prof. Hyeonwoo Yu

Graduate Coursework GPA: 4.17 / 4.5

Expected Graduation: August 2029
- 2017.03 – 2023.08

B.S. in Electrical Engineering

UNIST

Publications

2026

Real-Time LiDAR Gaussian Splatting SLAM via Geometry-Aware Covariance Coupling

SeungJun Tak*, Yewon Jeon*, Jaeik Hwang, Suk Min Hwang, Seongbo Ha, Hyeonwoo Yu

ECCV 2026 · Accepted · * Equal contribution

Real-time LiDAR-only SLAM with dense Gaussian mapping, coupling geometric covariance between tracking and mapping.

[Paper](#) [Code](#) [Project](#) [Video](#)

2026

LangGS-SLAM: Real-Time Language-Feature Gaussian Splatting SLAM

Seongbo Ha, Sibaek Lee, Kyungsu Kang, Joonyeol Choi, Seungjun Tak, Hyeonwoo Yu

arXiv:2602.06991 · Preprint

[Paper](#)

2026

Robust Localization in Changing Logistics Environments

변화하는 물류 환경에 강건한 위치 추정 기법

SeungJun Tak, Hyeonwoo Yu

KRoC 2026 · Published

2025

LEGO-SLAM: Language-Embedded Gaussian Optimization SLAM

Sibaek Lee, Seongbo Ha, Kyeongsu Kang, Joonyeol Choi, Seungjun Tak, Hyeonwoo Yu

arXiv:2511.16144 · Preprint

[Paper](#) [Code](#) [Project](#)

2025

SLAM in Logistics Environments Using Pillar Feature Points

기둥 특징점을 활용한 물류창고 환경 SLAM

SeungJun Tak, Hyeonwoo Yu

KRoC 2025 · Published

Research Projects

- 2026.07 – 2027.04

Real-Time 3D RGB-D SLAM System for Humanoid Robots and Low-Profile AMRs

LG Electronics, HS · Graduate Researcher · Ongoing

Developing **real-time SLAM** for humanoid robots and low-profile AMRs, focusing on robust localization and 3D mapping.

- 2026.04 – 2026.12

Development of a SLAM-Drone-Based Cart Monitoring System

LG PRI · Graduate Researcher

Optimized SLAM for aerial drones and developed **loop closure** for robust trajectory estimation in cart monitoring.
- 2026.01 – 2026.11

Physical AI-Based Integrated Autonomous Navigation for AMRs

Samsung Electronics GTR · Graduate Researcher

Aligned **CAD floor plans with reconstructed 3D maps** and optimized localization and mapping for AMR deployment.
- 2025.05 – 2026.05

Real-Time High-Resolution 3D RGB-D SLAM for Embedded Systems in Changing Environments

LG Electronics, HS · Graduate Researcher

Developed localization on **Gaussian Splatting maps** and MAST3R-based place recognition for robust global initialization and loop-closure detection.
- 2024.08 – 2025.03

Sensor Fusion 3D LiDAR Mapping and Map-Based Localization for AMRs

Mobyus · Graduate Researcher

Designed **pole-based geometric features** for robust LiDAR SLAM and map-based localization in repetitive warehouses.
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Teaching Experience

- 2025.09 – 2025.12

Teaching Assistant — Reinforcement Learning

Sungkyunkwan University
- 2025.03 – 2025.06

Teaching Assistant — Estimation Theory and Nonparametric Bayesian Methods

Sungkyunkwan University
- 2023.09 – 2024.02

Teaching Assistant — 3D SLAM Techniques and Adaptive Perception Technologies Using Unsupervised Learning

UNIST
- 2023.04 – 2023.07

Teaching Assistant — Kyungnam AI Novatus Academy
- 2022.08

Teaching Assistant — LG DX Intensive Course

Privacy-Preserving Machine Learning
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Technical Skills

Programming	C++, Python, CUDA
Robotics & SLAM	ROS/ROS 2, LiDAR SLAM, Visual SLAM, Sensor Fusion, G-ICP, Loop Closure, Localization
3D Vision & Mapping	Gaussian Splatting, 3D Reconstruction, Point Cloud Processing, Place Recognition
Libraries & Frameworks	PyTorch, GTSAM, PCL, Open3D, Eigen
Development	Linux, Git, Docker, CMake